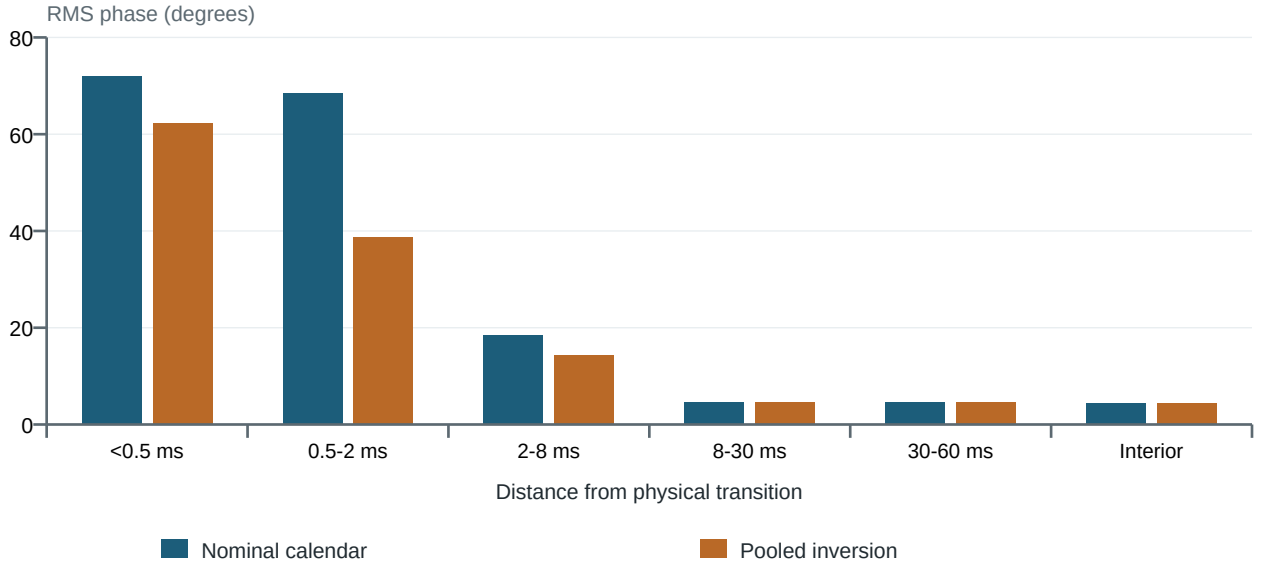Complete output: phase error by physical distance from a hop